Chhatrapati Shahu Ji Maharaj University

Project Synopsis

on

AI Book – Digital Notes System

Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Computer Application

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ABSTRACT

Ask any college student where they keep their notes, and you will likely get a different answer every time – a WhatsApp group here, a friend's phone gallery there, a stack of photocopied pages that never quite makes it to exam week in one piece. Important study material gets scattered across chats, drives, and printouts, and a lot of time that should go into actual learning ends up going into simply finding the right notes. AI Book – Digital Notes System was born out of this everyday problem. It is a web-based platform that gives a college one single, organized place to store and access academic notes, so that students spend less time searching and more time studying.

At its core, the system keeps things simple: administrators and faculty upload notes as PDFs, neatly organized by subject and semester, and students can search, view, and download exactly what they need in a few clicks. But what makes AI Book more than just a file-sharing tool is the Artificial Intelligence woven into it. Long, dense notes are automatically summarized into shorter, easy-to-revise versions. Students can simply ask a question and get an answer drawn directly from their own course material, almost like having a study partner who has already read everything. And when it's time to test their understanding, the system can generate topic-wise quizzes on its own, turning passive reading into active revision.

Taken together, these features are meant to make studying feel a little less overwhelming and a lot more organized. By cutting down the manual effort of sharing notes and adding a genuinely helpful AI layer on top, AI Book aims to make quality study material more accessible, and to give students a smarter, more supportive way to prepare – whenever and wherever they need it.

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1. INTRODUCTION

1.1 Objective

The primary objective of this project is to develop a centralized, AI-assisted platform for managing and accessing academic study materials. The system aims to:

- Provide a single, organized platform for uploading and accessing subject-wise and semester-wise notes.
- Reduce the manual effort involved in sharing notes across classes and batches.
- Use Artificial Intelligence to summarize lengthy PDF notes into concise, easy-to-revise content.
- Allow students to ask questions and get answers based on the content of the uploaded notes.
- Generate topic-wise quizzes automatically to help students self-assess their understanding.
- Improve accessibility of educational resources for students at any time and from any device.

1.2 Scope

This project focuses on building a secure, role-based web platform for notes management enhanced with AI-driven learning tools. The scope includes:

- Admin/Faculty Module: Secure upload, categorization, and management of notes by subject and semester.
- Student Module: Search, view, and download notes with a simple, intuitive interface.
- AI Summarization: Automatic generation of concise summaries from uploaded PDF notes.
- AI Question Answering: A chat-based assistant that answers student queries using the content of the uploaded notes.
- AI Quiz Generation: Automatic creation of topic-wise quizzes for self-assessment based on notes content.
- Access Control: Role-based authentication to differentiate between administrators, faculty, and students.

2. SYSTEM REQUIREMENTS

2.1 Software Requirements

- Programming Languages: JavaScript (ES6)
- Frontend: HTML5, CSS3, JavaScript (ES6)
- Backend Runtime & Framework: Node.js, Express.js
- Database: MySQL
- AI Technologies: Google Gemini API, pdf-parse
- Authentication & Security: JSON Web Token (JWT), bcrypt, dotenv
- File Upload: Multer
- Development Tools: VS Code, Postman, Git & GitHub, MySQL Workbench

2.2 Hardware Requirements

- Processor: Intel i5/i7 or AMD Ryzen equivalent
- RAM: Minimum 8GB (16GB recommended for running AI/NLP models)
- Storage: At least 20GB free space (for application, database, and cached PDF files)
- Internet Connectivity: Required for AI API calls and cloud storage access

3. MODULES

3.1 Module Names

- Authentication & User Management Module
- Notes Upload & Management Module
- Notes Search & Access Module
- AI Summarization Module
- AI Question-Answering Module
- AI Quiz Generation Module

3.2 Module Description

Authentication & User Management Module: Handles secure login and registration for admins, faculty, and students. Manages role-based access so that only authorized users can upload or manage notes.

Notes Upload & Management Module: Allows administrators and faculty to upload PDF notes tagged by subject, semester, and topic. Supports editing, replacing, and deleting existing notes.

Notes Search & Access Module: Enables students to search notes by subject, semester, or keyword, and to view or download the required PDF files through a clean, organized interface.

AI Summarization Module: Processes uploaded PDF notes using NLP techniques to generate concise, easy-to-read summaries, helping students revise large volumes of content quickly.

AI Question-Answering Module: Uses a language model to answer student questions based strictly on the content of the uploaded notes, acting as a study assistant available at any time.

AI Quiz Generation Module: Automatically generates topic-wise multiple-choice or short-answer quizzes from the notes content, allowing students to self-assess their understanding of each topic.

4. APPLICATION ARCHITECTURE

1. Data Collection & Storage: Administrators and faculty upload subject-wise and semester-wise PDF notes through the web interface. Files are stored in cloud/file storage, and metadata (subject, semester, topic, uploader) is stored in the database.

2. Notes Processing & AI Pipeline: Uploaded PDFs are parsed and their text extracted. The extracted content is passed through NLP models to generate summaries, build a searchable knowledge base for question-answering, and create topic-wise quiz questions.

3. Student Interaction Layer: Students interact with the system through a web dashboard to search and download notes, request AI-generated summaries, ask questions to the AI assistant, and attempt auto-generated quizzes.

4. Backend & API Layer: A REST API built using Node.js and Express.js handles requests between the frontend and backend, manages JWT-based authentication, and routes AI-related requests to the summarization, Q&A, and quiz-generation services.

5. Continuous Improvement: Usage data such as quiz performance and frequently asked questions can be analyzed to improve content organization and identify topics that need clearer notes.

5. TECHNOLOGY USED

5.1 Frontend Technologies

Technology	Purpose
HTML5	To design the structure of web pages.
CSS3	To create an attractive and responsive user interface.
JavaScript (ES6)	To make the web pages interactive and communicate with backend APIs using Fetch API or AJAX.

5.2 Backend Technologies

Technology	Purpose
Node.js	JavaScript runtime used to build the backend server.
Express.js	Framework used to create RESTful APIs and manage routing, middleware, and HTTP requests.

5.3 Database

Technology	Purpose
MySQL	Stores administrator, student, subject, and notes information securely in a relational database.

5.4 AI Technologies

Technology	Purpose
Google Gemini API	Generates summaries, answers questions from uploaded notes, and creates quizzes using AI.
pdf-parse	Extracts text from uploaded PDF files so the AI can process the content.

5.5 Authentication & Security

Technology	Purpose
JSON Web Token (JWT)	Provides secure user authentication and authorization.
bcrypt	Encrypts user passwords before storing them in the database.
dotenv	Stores sensitive information such as database credentials and API keys securely in environment variables.

5.6 File Upload Technology

Technology	Purpose
Multer	Handles uploading and storing PDF files on the server.

5.7 APIs Used

REST API: The project follows the REST (Representational State Transfer) architecture to enable communication between the frontend and backend.

Examples of APIs: Authentication APIs, Notes Upload/Retrieval APIs, AI Summarization API, AI Question-Answering API, and AI Quiz Generation API.

5.8 Development Tools

Software	Purpose
Visual Studio Code (VS Code)	Source code editor used for project development.
Postman	Testing REST APIs during backend development.
Git	Version control system used to track source code changes.
GitHub	Cloud repository used for source code management and collaboration.
MySQL Workbench	Database design, query execution, and management.
Google Chrome	Testing the web application and APIs.

5.9 NPM Packages Used

Package	Purpose
express	Creates the web server and REST APIs.
mysql2	Connects the application to the MySQL database.
jsonwebtoken	Generates and verifies JWT tokens.
bcrypt	Hashes passwords securely.
multer	Uploads PDF files.
dotenv	Loads environment variables from the .env file.
cookie-parser (Optional)	Parses cookies if required in future enhancements.
pdf-parse	Extracts text from uploaded PDF files.
@google/generative-ai	Connects the application to the Google Gemini API for AI features.

6. REFERENCES

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